Laborator 5

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① 3 trăgători

Notăm cu A_i evenimentul ca trăgătorul i nimerește ținta.

$$\mathbb{P}(A_1)=\frac{2}{3}, \quad \mathbb{P}(A_2)=\frac{3}{4}, \quad \mathbb{P}(A_3)=\frac{4}{5}$$

Notăm cu X evenimentul ca ținta a fost atinsă, B evenimentul ca ținta să fie atinsă o singură dată

$\mathbb{P}(X)=0,91$ $\mathbb{P}(B)=0,15$

a) $\mathbb{P}(A_1|X)=\frac{\mathbb{P}(A_1\cap X)}{\mathbb{P}(X)}=\frac{\mathbb{P}(A_1\cap(A_1\cup A_2\cup A_3))}{\mathbb{P}(X)}=\frac{\mathbb{P}(A_1)}{\mathbb{P}(X)}=\frac{\frac{2}{3}}{0,91}\simeq 0,73$

d) $\mathbb{P}(A_2|B)=\frac{\mathbb{P}(A_2\cap B)}{\mathbb{P}(B)}=\frac{\mathbb{P}(A_2\cap(A_1\cap\bar{A}_2\cap\bar{A}_3\cup\bar{A}_1\cap A_2\cap\bar{A}_3\cup\bar{A}_1\cap\bar{A}_2\cap A_3))}{\mathbb{P}(B)}$

$$=\frac{\mathbb{P}(\emptyset\cup\emptyset\cup\bar{A}_1\cap A_2\cap\bar{A}_3)}{\mathbb{P}(B)}$$

$$=\frac{\mathbb{P}(\bar{A}_1\cap A_2\cap\bar{A}_3)}{\mathbb{P}(B)}$$

$$\overset{A_1,A_2,A_3}{\underset{ind}{=}}\frac{\mathbb{P}(\bar{A}_1)\cdot\mathbb{P}(A_2)\cdot\mathbb{P}(\bar{A}_3)}{\mathbb{P}(B)}$$

$$=\frac{\frac{1}{\cancel{3}}\cdot\frac{\cancel{3}}{4}\cdot\frac{1}{5}}{0,15}=\frac{\frac{1}{20}}{0,15}\simeq 0,33$$

⑩ F2

100 calculatoare < 30 F1, 50 F2, 20 F3

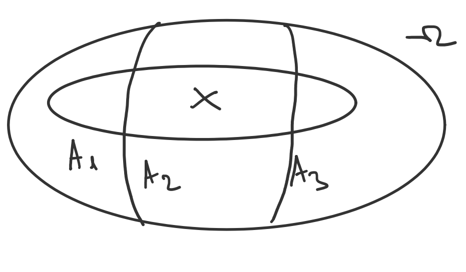
2%	$\mathbb{P}(A_1)=30\%$	$\mathbb{P}(X\|A_1)=2\%$
4%	$\mathbb{P}(A_2)=50\%$	$\mathbb{P}(X\|A_2)=4\%$
5%	$\mathbb{P}(A_3)=20\%$	$\mathbb{P}(X\|A_3)=5\%$

Notăm cu A_i evenimentul că un calculator provine de la F_i, $i=\overline{1,3}$

X evenimentul că un calculator se defectează

a) $\mathbb{P}(X)=?$

$$\mathbb{P}(X)=\mathbb{P}(X\cap\Omega)=\mathbb{P}(X\cap(A_1\cup A_2\cup A_3))\overset{A_1,A_2,A_3\ inc}{\downarrow =}\underbrace{\mathbb{P}(X\cap A_1\cup X\cap A_2\cup X\cap A_3)}_{inc}$$



$$=\mathbb{P}(X\cap A_1)+\mathbb{P}(X\cap A_2)+\mathbb{P}(X\cap A_3)$$

$\mathbb{P}(X|A_1)=\frac{\mathbb{P}(X\cap A_1)}{\mathbb{P}(A_1)}\Rightarrow\mathbb{P}(X\cap A_1)=\mathbb{P}(A_1)\cdot\mathbb{P}(X|A_1)$

$\mathbb{P}(X|A_2)=\frac{\mathbb{P}(X\cap A_2)}{\mathbb{P}(A_2)}\Rightarrow\mathbb{P}(X\cap A_2)=\mathbb{P}(A_2)\cdot\mathbb{P}(X|A_2)$

$\mathbb{P}(X|A_3)=\frac{\mathbb{P}(X\cap A_3)}{\mathbb{P}(A_3)}\Rightarrow\mathbb{P}(X\cap A_3)=\mathbb{P}(A_3)\cdot\mathbb{P}(X|A_3)$

$$\mathbb{P}(X)=\mathbb{P}(X|A_1)\cdot\mathbb{P}(A_1)+\mathbb{P}(X|A_2)\cdot\mathbb{P}(A_2)+\mathbb{P}(X|A_3)\cdot\mathbb{P}(A_3)$$

$$=2\%\cdot30\%+4\%\cdot50\%+5\%\cdot20\%=\frac{6\cancel{0}}{1000\cancel{0}}+\frac{20\cancel{0}}{1000\cancel{0}}+\frac{10\cancel{0}}{1000\cancel{0}}\simeq 0,036$$

Obs! Un șir de ev $A_1, A_2, \ldots, A_n$ cu propr.:

1. $A_i\cap A_j=\emptyset$, $\forall i\neq j$, $i,j=\overline{1,n}$ (oricare două câte două)
2. $\bigcup_{i=1}^{n}A_i=\Omega$

formează o partiție a lui Ω.

Formula probabilității totale

Prob ∀ ev din Ω poate fi scrisă ca:

$$\mathbb{P}(X)=\sum_{i=1}^{3}\mathbb{P}(X|A_i)\cdot\mathbb{P}(A_i)$$ (caz particular problemă)

Tips & Tricks!
- care (știind că) - ev care condiționează
- să - ev de interes

b) $\mathbb{P}(A_2|X)=\frac{\mathbb{P}(A_2\cap X)}{\mathbb{P}(X)}$

$$=\frac{\mathbb{P}(X|A_2)\cdot\mathbb{P}(A_2)}{\mathbb{P}(X)}$$

$$=\frac{4\%\cdot50\%}{0,036}=\frac{\frac{20\cancel{0}}{1000\cancel{0}}}{0,036}\simeq 0,55$$

c) $\mathbb{P}(X|(A_1\cup A_3))=\frac{\mathbb{P}(X\cap(A_1\cup A_3))}{\mathbb{P}(A_1\cup A_3)}\overset{A_1,A_3\ inc}{\downarrow =}\frac{\overbrace{\mathbb{P}(X\cap A_1\cup X\cap A_3)}^{inc}}{\underset{inc}{\mathbb{P}(A_1\cup A_3)}}=$

$$=\frac{\mathbb{P}(X\cap A_1)+\mathbb{P}(X\cap A_3)}{\mathbb{P}(A_1)+\mathbb{P}(A_3)}=\frac{\mathbb{P}(X|A_1)\,\mathbb{P}(A_1)+\mathbb{P}(X|A_3)\cdot\mathbb{P}(A_3)}{\mathbb{P}(A_1)+\mathbb{P}(3)}$$

$$=\frac{2\%\cdot30\%+5\%\cdot20\%}{30\%+20\%}=\frac{\frac{6\cancel{0}}{1000\cancel{0}}+\frac{10\cancel{0}}{1000\cancel{0}}}{\frac{5\cancel{0}}{10\cancel{0}}}=\frac{\frac{16}{1000}}{\frac{5}{10}}=$$

$$=\frac{16}{100\cancel{0}}\cdot\frac{1\cancel{0}}{5}==\frac{16}{500}=0,032$$

d) $\mathbb{P}((A_1\cup A_2)|\bar{X})=\frac{\mathbb{P}((A_1\cup A_2)\cap\bar{X})}{\mathbb{P}(\bar{X})}\overset{A_1,A_2\ inc}{\downarrow =}\frac{\overbrace{\mathbb{P}(A_1\cap\bar{X}\cup A_2\cap\bar{X})}^{inc}}{\mathbb{P}(\bar{X})}$

$$=\frac{\mathbb{P}(A_1\cap\bar{X})+\mathbb{P}(A_2\cap\bar{X})}{\mathbb{P}(\bar{X})}$$

$$=\frac{\mathbb{P}(\bar{X}|A_1)\cdot\mathbb{P}(A_1)+\mathbb{P}(\bar{X}|A_2)\cdot\mathbb{P}(A_2)}{\mathbb{P}(\bar{X})}$$

$$=\frac{(1-\mathbb{P}(X|A_1))\cdot\mathbb{P}(A_1)+(1-\mathbb{P}(X|A_2))\cdot\mathbb{P}(A_2)}{1-\mathbb{P}(X)}$$

$$=\frac{98\%\cdot30\%+96\%\cdot50\%}{1-0,036}$$

Obs

$\mathbb{P}(\bar{A}|B)=1-\mathbb{P}(A|B)$

$\mathbb{P}(A|\bar{B})\neq 1-\mathbb{P}(A|B)$

$$\boxed{\text{Bayes}: \mathbb{P}(A|B)\cdot\mathbb{P}(B)=\mathbb{P}(B|A)\cdot\mathbb{P}(A)}$$

$$\mathbb{P}(A|\bar{B})=\frac{\mathbb{P}(\bar{B}|A)\cdot\mathbb{P}(A)}{\mathbb{P}(\bar{B})}=\frac{(1-\mathbb{P}(B|A))\,\mathbb{P}(A)}{\mathbb{P}(\bar{B})}$$

② Calculați de câte ori tb aruncat un zar a.î. prob ca pe fața superioară a acestuia să apară numărul 6 cel puțin o dată să fie de cel puțin 0,9.

Not A_i ev ca pe fața superioară să apară 6 la aruncarea i, $i=\overline{1,k}$

k - nr de aruncări necesare

Se subînțelege că aruncările sunt independente.

K = ?

$$\mathbb{P}(A_1\cup A_2\cup\ldots\cup A_k)=\mathbb{P}\left(\bigcup_{i=1}^{k}A_i\right)\geq 0,9\Rightarrow$$

$$\Rightarrow 1-\mathbb{P}\left(\bigcup_{i=1}^{k}A_i\right)\leq 0,1\Leftrightarrow$$

$$\Leftrightarrow\mathbb{P}\left(\overline{\bigcup_{i=1}^{k}A_i}\right)\leq 0,1\Leftrightarrow\mathbb{P}\left(\bigcap_{i=1}^{k}\bar{A}_i\right)\leq 0,1\overset{A_i\ ind.}{\Leftrightarrow}$$

$$\Leftrightarrow\mathbb{P}\left(\prod_{i=1}^{k}\bar{A}_i\right)\leq 0,1\Leftrightarrow\left(\frac{5}{6}\right)^{k}\leq 0,1\ \Big|\ \lg$$

$$\Leftrightarrow k\lg\frac{5}{6}\leq\lg 0,1\Leftrightarrow k\geq\frac{\lg 0,1}{\lg\frac{5}{6}}=\frac{1}{0,07}\simeq 14,28$$

$\Leftrightarrow k\geq 14,28$, deci

$K\in\{15,16,\ldots\}$